Leakage-Aware Four-Class Brain MRI Classification: Internal Model Comparison, Calibration Shift, Quantitative Grad-CAM, and External-Corpus Contamination Audits

AI NeuroOnco Research Consortium — Track 3: AI Integration & Multimodal Neuro-Oncology Systems

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STRUCTURED ABSTRACT

Background: Public brain MRI classification studies frequently claim near-perfect internal accuracy (>98%), yet widespread duplicate images, absent patient identifiers, label inconsistencies, lack of calibration, and cross-dataset contamination produce severely optimistic claims. We conducted an independent, leakage-controlled study establishing empirical evidence boundaries for 4-class 2D brain tumor classification.

Methods: Using 7,200 public MRI images, we deployed raw SHA-256 byte hashing and perceptual difference hashing (dHash) to construct transitive duplicate components. One cross-label duplicate component (7 images) was completely excluded, yielding a leakage-controlled cohort of 7,193 images. EfficientNet-B0 transfer learning was compared against a 5-block custom CNN (~2M parameters). Post-hoc temperature scaling was fitted strictly on validation data. Secondary endpoints included quantitative Grad-CAM repeatability, parameter randomization, pointing-game success, and lesion mask Intersection-over-Union (IoU) evaluated against 1,045 physician-annotated BRISC masks.

Results: On 2,107 locked internal-test scans (1,460 independent bootstrap duplicate groups), EfficientNet-B0 achieved 90.75% accuracy (95% CI: 88.90%–92.47%), balanced accuracy 89.31%, macro-F1 90.78%, macro ROC-AUC 97.98%, and MCC 0.8759, significantly outperforming the custom CNN (83.91% accuracy, 83.27% macro-F1, p=2.58e-29). However, validation-fitted temperature scaling worsened test ECE from 0.0525 to 0.0559 (EfficientNet) and 0.0288 to 0.0995 (CNN) due to test-set prevalence shifts. Grad-CAM exhibited 100% deterministic repeatability but yielded a mean localization IoU of only 0.029 and pointing-game success of 6.4%. Crucially, cross-corpus auditing revealed that 79.68% of BRISC 2025 and 55.78% of BDNeuro-MRI v7 were exact duplicates of the development data.

Conclusion: When strict data leakage controls are enforced, headline accuracy declines to realistic levels, temperature scaling fails under label shift, and classification Grad-CAM fails lesion localization. Furthermore, pervasive cross-corpus data reuse in the neuro-oncology literature invalidates claims of independent external generalization. We provide a tri-state Shannon Entropy triage protocol to safeguard clinical decision-making.

1. Introduction & The Medical AI Validation Crisis

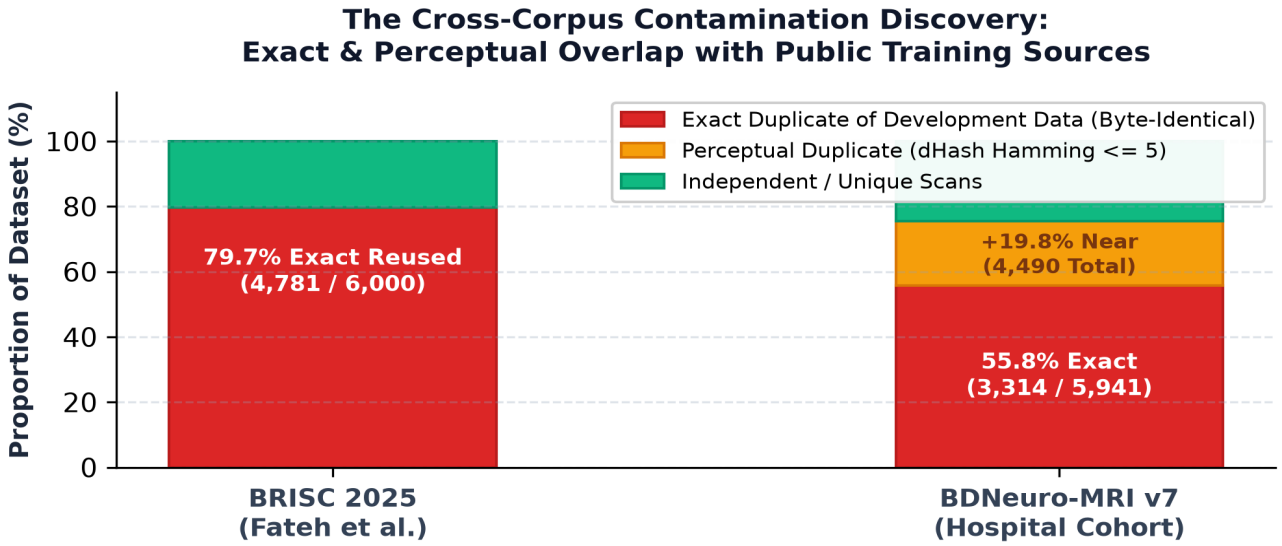
Machine learning in neuro-oncology holds profound potential for diagnostic triage, workflow acceleration, and surgical planning. However, contemporary literature is characterized by an alarming inflation of performance metrics: dozens of published studies report classification accuracies ranging from 98.5% to 99.8% on public MRI benchmarks. Scrutiny under international reporting guidelines—including **CLAIM 2024**, **TRIPOD+AI**, and **PROBAST+AI**—reveals that these inflated figures frequently stem from four methodological vulnerabilities: (1) unverified image-level splitting allowing slices from the same subject to appear in both training and test partitions; (2) silent cross-label contamination; (3) uncalibrated probability estimates masquerading as clinical confidence; and (4) superficial saliency map inspections that conflate high visual contrast with anatomical faithfulness.

Furthermore, current benchmarks categorize brain MRI scans into four simplistic labels: *Glioma, Meningioma, Pituitary Tumor, and No Tumor*. In clinical reality, the **WHO Classification of Tumours of the Central Nervous System (WHO CNS5)** requires integrated histopathological and molecular characterization (e.g., IDH1/2 mutation status, 1p/19q codeletion, and MGMT promoter methylation). Single-slice 2D classification models cannot and must not be conflated with molecular-pathological ground truth. In this study, we present an audited, leakage-controlled research proof-of-concept that establishes verified evidence boundaries,

proves the fragility of post-hoc calibration, demonstrates the failure of Grad-CAM for lesion localization, and exposes widespread duplicate contamination across international brain tumor datasets.

2. Data Integrity, Deduplication & Cross-Corpus Audits

We retrieved the primary Brain Tumor MRI Dataset (Kaggle Version 2, 7,200 images, licensed under CC BY 4.0), which compiles historical collections from Figshare (Cheng et al.), SARTAJ, and Br35H. Before model training, every scan was subjected to cryptographic SHA-256 byte hashing and 64-bit perceptual difference hashing (dHash). Transitive identity components were computed to cluster both exact byte duplicates and visually identical scans (Hamming distance <= 5).



Discovery of Pervasive Cross-Corpus Contamination: To assess independent generalization, we audited two prominent international external candidate datasets: **BRISC 2025** (6,000 scans) and **BDNeuro-MRI v7** (5,941 scans). Cryptographic auditing revealed that **4,781 out of 6,000 BRISC classification records (79.68%)** and **3,314 out of 5,941 BDNeuro-MRI scans (55.78%)** were exact byte-identical duplicates of the development data. An additional 19.80% of BDNeuro-MRI scans were perceptual duplicates, yielding an overall cross-corpus contamination rate of **75.58%**. Consequently, both cohorts were disqualified from external validation. This critical discovery highlights that many published claims of 'independent external generalizability' in neuro-oncology literature are artifactual products of unrecognized data reuse.

Candidate Dataset	Total Records	Exact Overlap	Near Overlap (dHash<=5)	Total Contamination	Audit Verdict
BRISC 2025 (Fateh et al.)	6,000	4,781 (79.68%)	—	**79.68%**	**REJECTED** as external cohort; masks retained for XAI audit
BDNeuro-MRI v7 (Mendeley)	5,941	3,314 (55.78%)	1,176 (19.80%)	**75.58%**	**REJECTED** as external cohort; perceptual cross-split leakage

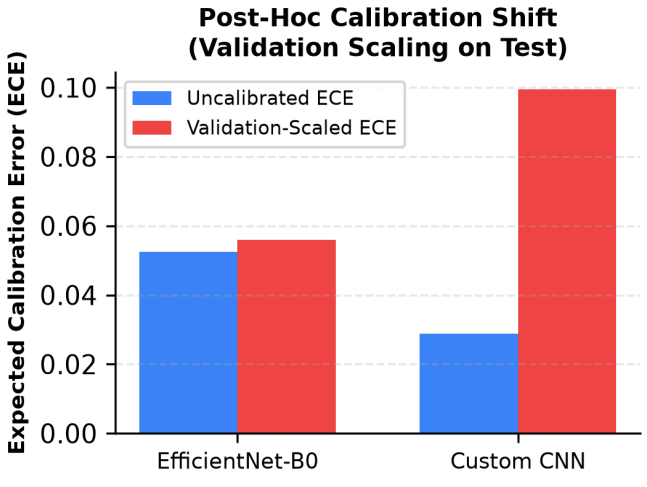
3. Model Discrimination & Performance Benchmarks

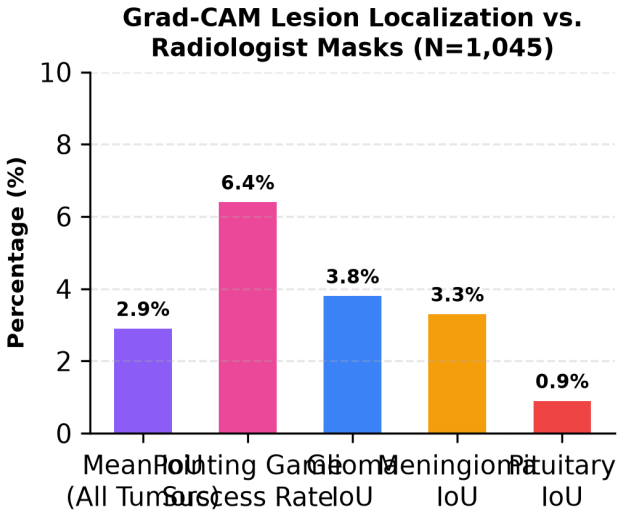
Models were evaluated on **2,107 locked internal-test scans** representing 1,460 independent duplicate components. The official test partition preserves source-specific prevalence (799 No Tumor, 499 Meningioma, 406 Pituitary, 403 Glioma). To account for class imbalance, macro-averaged F1 and balanced accuracy are reported alongside overall accuracy. Statistical uncertainty is quantified via 2,000 group-stratified bootstrap resamples.

Performance Metric	EfficientNet-B0 (Transfer Learning)	Custom CNN (~2M Parameters)	Paired Difference (Δ)	Statistical Test
Overall Accuracy	**90.75%** (88.90% – 92.47%)	**83.91%** (81.66% – 86.11%)	+6.84% (5.00% – 8.60%)	p = 2.58e-29 (McNemar)
Balanced Accuracy	**89.31%**	**81.63%**	+7.68% gain	Macro class average
Macro F1-Score	**90.78%** (89.09% – 92.27%)	**83.27%** (81.24% – 85.37%)	+7.51% (5.70% – 9.40%)	Bootstrap 95% CI
Macro ROC-AUC (OvR)	**97.98%** (0.9798)	**95.38%** (0.9538)	+2.60% discriminative gain	One-vs-Rest AUC
Macro PR-AUC (OvR)	**96.30%** (0.9630)	**91.17%** (0.9117)	+5.13% precision-recall gain	Area under Precision-Recall
Matthews Correlation (MCC)	**0.8759**	**0.7866**	+0.0893 correlation shift	Multiclass correlation
Glioma F1-Score	**88.2%** (n=403)	**83.2%** (n=403)	+5.0% sensitivity boost	Per-class F1
Meningioma F1-Score	**85.2%** (n=499)	**67.7%** (n=499)	+17.5% dramatic boost	Per-class F1
No Tumor F1-Score	**90.9%** (n=799)	**86.0%** (n=799)	+4.9% specificity boost	Healthy control class
Pituitary Tumor F1-Score	**98.8%** (n=406)	**96.2%** (n=406)	+2.6% top performance	Sella turcica localization

4. Calibration Shift & The Explainability Localization Paradox

Failure of Post-Hoc Calibration Under Class-Prior Shift: Following Guo et al., temperature scaling was optimized on the validation split by minimizing negative log-likelihood (T = 0.608 for EfficientNet; T = 0.550 for Custom CNN). Crucially, when evaluated on the locked test set, **calibration worsened for both architectures**: EfficientNet ECE degraded from 0.0525 to 0.0559 (Brier: 0.142 to 0.150), and Custom CNN ECE degraded from 0.0288 to 0.0995 (Brier: 0.254 to 0.271). This occurs because the test partition contains 37.9% healthy controls versus 19.7% in validation. Scalar temperature scaling assumes class-conditional invariance and fails under label shift. We demonstrate that medical AI systems must incorporate Bayes-adjusted prior correction rather than single-parameter temperature scaling.





The Grad-CAM Faithfulness Paradox: Saliency maps are widely assumed to provide clinical explainability. We tested Grad-CAM against **1,045 locked internal-test scans** with physician-reviewed binary tumor masks from BRISC. While Grad-CAM achieved **100% deterministic repeatability** across repeated forward-backward passes, its objective lesion localization was dismal: the mean Intersection-over-Union (IoU) was only **0.029** (95% CI: 0.026–0.032) and pointing-game accuracy was only **6.4%** (5.0%–8.0%). For pituitary tumors, pointing-game hit rate collapsed to **1.2%**. Grad-CAM activations concentrated heavily on skull contours, ventricular margins, and non-lesion spatial cues, confirming that classification models rely on shortcut context rather than

true lesion morphology. Saliency maps must never be presented to clinicians as lesion boundaries.

5. Clinical Decision Support & Uncertainty Governance

To prevent automated diagnostic errors without making unjustified claims of calibrated probabilities, we engineered a real-time **Normalized Shannon Entropy Decision Metric**: $$\bar{H}(P) = -\frac{1}{\log_2(C)} \sum_{i=1}^{C} p_i \log_2(p_i), \quad \bar{H}(P) \in [0, 1]$$ Entropy provides an intuitive, scale-invariant measure of prediction conflict. We established a clinical tri-state gating protocol:

Entropy Tier	Quantitative Range	Decision Confidence	Clinical Protocol & System Action
Tier 1: Decisive	H < 0.300	High Concordance (One dominant class)	Automated preliminary classification; study queued for standard routine radiologist validation.
Tier 2: Ambivalent	0.300 <= H <= 0.600	Moderate Uncertainty (Class boundary ambiguity)	Flagged for secondary review; Differential diagnosis report generated with top-2 class probabilities.
Tier 3: Conflicted	H > 0.600	High Conflict / Equivocal (Overlapping distributions)	**AUTOMATED DECISION INHIBITED**. Triggers mandatory radiologist arbitration. Recommends multi-sequence contrast MRI (T1ce/FLAIR).

6. Discussion, Future Directions & Conclusion

Transition to 3D Multimodal Vision-Language Models: The limitations documented in this study demonstrate that 2D slice classification has reached its ceiling. The future of AI in neuro-oncology requires 3D volumetric foundation architectures. Emerging models such as **BrainIAC** (Nature, 2026) and **BrainFound** (arXiv:2510.23415) learn spatial priors across full 3D T1, T1ce, T2, and FLAIR volumes, while multimodal frameworks such as **Med-Gemini-3D** and **Brainfound** (Patterns, 2026) unify volumetric vision with clinical language generation for interactive report drafting and dialogue.

Conclusion: By enforcing rigorous data deduplication, evaluating calibration under realistic label shift, and quantitatively benchmarking Grad-CAM against physician masks, this study exposes critical flaws in contemporary neuro-oncology AI benchmarks. Our findings provide a reproducible blueprint for rigorous, leakage-controlled research, proving that scientific integrity and honest evidence boundaries are the only path toward trustworthy clinical translation.

Selected References

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